FFGFT in the Information Formalism

Mapping, Projection, and Residuals

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Contents

Preliminary Note

This document examines whether and how the geometric description of FFGFT can be mapped into an information formalism. The occasion is the discussion with Onur Teker (UIFT) on whether geometry or information is the more fundamental primitive.

The central thesis: **FFGFT is fully mappable into an information formalism** — just as Doc. 230 showed for the Hilbert space formalism. However, this mapping is *not an equivalence*: information is lost under projection (winding numbers, θ_4 phase, non-closure ε). The mapping is a projection π, not an isomorphism.

Methodological consequence: That FFGFT can be mapped into an information formalism does not prove that information is prior. It shows that both descriptions are projections of the same structure — with different residuals.

.1 Basic Structure of the Mapping

The geometric state space in FFGFT

In FFGFT the fundamental state space is the T^4 torus with winding numbers $(n_\theta, n_\varphi) \in \mathbb{Z}^2$ and phase $\theta_4 \in [0, 2\pi)$. A physical state Ψ is fully characterised by:

$$\Psi = \left(n_\theta,\ n_\varphi,\ \theta_4,\ \varepsilon\right) \tag{243.1}$$

where $\varepsilon \approx 0.001$ is the non-closure (Doc. 193).

Explicit definition of π_I

The projection π_I is defined as:

$$\pi_I:\ \Psi(n_\theta, n_\varphi, \theta_4, \varepsilon) \longmapsto \left(I(n_\theta, n_\varphi),\ \mathrm{sgn}(\theta_4),\ \langle\varepsilon\rangle_{\mathrm{therm}}\right) \tag{243.2a}$$

where $I = \log_2 |\mathscr{W}|$ is the bit count of admissible winding configurations, $\mathrm{sgn}(\theta_4)$ is the binary collapse of the continuous phase, and $\langle\varepsilon\rangle_{\mathrm{therm}}$ is the non-closure appearing only as a thermal expectation value.

Important: $|\mathscr{W}|$ must be finite for I to be well-defined. The T⁴ geometry provides this bound via the maximum admissible winding number from the Planck-scale constraint $n_{\mathrm{max}} \sim 1/\xi$ (see Doc. 009). Without this geometric bound, I diverges — demonstrating that the information count *presupposes* the geometric state space.

Information-theoretic representation

The winding state (n_θ, n_φ) defines a finite set of N distinguishable configurations. The information content of a state is:

$$I = \log_2 N \quad \text{bits} \tag{243.2}$$

This is Boltzmann's formula in information-theoretic language: $S = k_B \ln N$ corresponds to $I = \log_2 N$ bits with appropriate scaling. The mapping:

$$(n_\theta, n_\varphi) \longmapsto I(n_\theta, n_\varphi) = \log_2 |\mathscr{W}| \tag{243.3}$$

where $\mathscr{W}$ is the set of admissible winding configurations defined by the Z_3 topology of T^4.

What is lost in the mapping

The information-theoretic projection π_I discards:

- The **continuous phase** θ_4 — collapsed to a binary distinction (phase present / absent)
- The **individual winding numbers** (n_θ, n_φ) — only their ratio $n_\varphi/n_\theta = 2s$ (spin) survives as a quantum number
- The **non-closure** ε — it appears in the information representation as "noise", not as a structural residual

This is the *residual* of the projection π_I: $R_I = \{\theta_4, (n_\theta, n_\varphi)_{\text{full}}, \varepsilon\}$.

Excursus: Extended Information Formalisms

The limitations of π_I are not absolute. Different information formalisms have different residuals:

Formalism	θ_4	(n_θ, n_φ)	ε	Residual
Classical-discrete (Shannon)	→1 bit	→spin	→noise	$\{\theta_4, (n_\theta, n_\varphi)_{\text{full}}, \varepsilon_{\text{geom}}\}$
Continuous (diff. entropy)	Yes (density)	→spin	Part. (Fisher)	$\{(n_\theta, n_\varphi)_{\text{full}}\}$
Quantum information (QIT)	Yes (phase)	→spin	→decoherence	$\{(n_\theta, n_\varphi)_{\text{full}}, \varepsilon_{\text{geom}}\}$
Topological (TDA)	→bit	Yes (inv.)	→noise	$\{\theta_4, \varepsilon_{\text{geom}}\}$

Result: No information formalism preserves all three quantities $(\theta_4, (n_\theta, n_\varphi)_{\text{full}}, \varepsilon)$ simultaneously. The T⁴ geometry is the only formalism that contains all three structurally. The residual of π_I is therefore *formalism-relative* — which strengthens the core argument: which quantities are lost depends on the chosen information formalism, not on the geometry.

.2 Concrete Translation Table

FFGFT (geometric)	Information formalism	Residual
Winding number (n_θ, n_φ)	Discrete state, $\log_2 N$ bits	Individual values lost; only ratio preserved
Phase $\theta_4 \in [0, 2\pi)$	1 bit (phase present / absent)	Continuous information lost
Non-closure $\varepsilon \approx 0.001$	Thermodynamic cost: $k_B T \ln 2$ per bit erasure	Geometric origin of ε not visible
$\tilde{T} \cdot m = 1$ (geometric, nat. units)	$\tau \cdot (\mu c^2) = \hbar$ (thermodynamic, SI); equiv. $\tau \cdot \mu = \hbar/c^2$ (SI: kg s) or $\tau \cdot \mu = 1$ (nat. units $\hbar = c = 1$)	None — this relation is invariant under π_I; $\tau \cdot \mu$ not dimensionless in SI
$\xi = 4/30000$ (winding ratio)	$\xi = k_B T_{\text{CMB}} \ln 2/(m_e c^2)$ (thermodynamic)	None — ξ survives as bridge parameter
$\Delta\text{CHSH}(N) \approx 5 \times 10^{-4}$	Landauer cost per measurement: $\xi/(2\pi) \approx 2 \times 10^{-5}$	Integration scale lost (cumulative vs. elementary)
Particle masses from windings	$m = r \cdot \xi^p \cdot v$ (formula)	Topological origin not visible

.3 Mapping into a Neural Network

Motivation

A neural network is an information-processing system whose activation patterns represent discrete states. If FFGFT states are representable as activation patterns, this would be a concrete information-theoretic realisation of FFGFT.

Architecture

A recurrent neural network (RNN) can implement the ξ-field equation

$$\Psi(t) \;=\; G\big(\Psi(t - T_{\text{rev}}),\; \xi\big) \tag{243.4}$$

as a recursion formula. The activation patterns of the hidden layer correspond to winding states:

$$h_t = \sigma\big(W \cdot h_{t-1} + b\big) \tag{243.5}$$

$$\text{with} \quad W_{ij} \sim (n_\theta, n_\varphi)_{\text{coupling matrix}} \tag{243.6}$$

The weight matrix W encodes the topological couplings between winding modes — it is the information-theoretic counterpart of the T^4 geometry.

What the network cannot encode

- The **continuous phase** θ_4: discrete activation functions collapse it
- The **exact non-closure** ε: it appears as training noise, not as a structural property
- The **geometric origin** of ξ: the network learns ξ as a hyperparameter, not as a winding ratio

The neural network is therefore a realisation of the projection π_I: it correctly encodes the informationally accessible part of FFGFT, but with the residuals noted above.

.4 Relation to Doc. 230: Hilbert Space Translation

Doc. 230 showed that FFGFT is fully translatable into Hilbert space quantum mechanics — with the residual that winding numbers are lost.

The present document shows the same for the information formalism: FFGFT is mappable — with the residual that θ_4, individual winding numbers, and the geometric origin of ε are lost.

Target	Projection	Equiv.?	Main residual
QM / Hilbert space	π_{QM}	No	Winding numbers
SM (field theory)	π_{SM}	No	θ_4 phase
Information formalism	π_I	No	θ_4 + ε origin
Matrix algebra	π_{Mat}	No	Both winding numbers

In all cases: FFGFT generates the target representation as a projection. The target representation cannot fully reconstruct FFGFT. **The mapping direction is asymmetric.**

.5 Consequence for the Priority Debate

The mappability of FFGFT into the information formalism does *not* prove that information is the ontological prior. It shows:

1. Both descriptions — geometric and information-theoretic — are internally consistent
2. Both are projections of a common structure Ψ
3. The geometric description has a smaller residual: it loses less when translated into other formalisms
4. The parameter $\xi = 4/30000$ is the *bridge parameter* that is identical in both descriptions — as a winding ratio and as a thermodynamic constant $k_B T_{\text{CMB}} \ln 2/(m_e c^2)$

Note on ξ-invariance: The invariance of ξ under π_I is *numerical*, not semantic: as the number $4/30000$ it is preserved, but its meaning changes (winding ratio ↔ thermodynamic energy ratio). This dual meaning is precisely its bridge function.

Working hypothesis: The T⁴ geometry is not "more true" than the information formalism, but it is *derivationally richer*: more physical parameters follow from it without additional input. Whether this is an ontological statement or merely a methodological one remains open — deliberately, since physics operates at the level of the measurable.

.6 Is a Neural Network a Genuine Information-Processing System?

The functional view

At the functional level, a neural network is an information-processing system in Shannon's sense: it receives inputs (distinguishable states), transforms them through a sequence of operations, and produces outputs (distinguishable states). Each layer performs a distinction — between activated and non-activated nodes — and the composition of layers produces a hierarchy of distinctions. This satisfies Shannon's definition: $I = \log_2 N$ bits per resolved alternative.

A broader AI system — combining neural networks with memory, retrieval, symbolic reasoning, and language models — extends this further. It maintains structured state across queries, selects among possible responses, and generates outputs that distinguish between alternatives. In the functional sense, such a system processes information in a well-defined way.

The fundamental view: geometry as prior

At the fundamental level, a neural network is a *geometric structure*: a directed graph with weighted edges (the weight matrix W), nonlinear node functions (activation), and a fixed topology (architecture). This geometric structure is defined entirely *before* any information is processed.

The architecture — which nodes exist, how they are connected, what the weight matrix looks like — is a geometric object. The information processing (the transformation of inputs to outputs) is what happens when this geometric structure is *instantiated* on a physical substrate and evaluated.

This mirrors the FFGFT situation exactly:

- T⁴ geometry defines which states are admissible (analogous to: the weight matrix defines which transformations are admissible)
- Physical particles are the instantiated states (analogous to: the network's output is the instantiated result)

- The information content follows from the geometry (analogous to: the number of distinguishable outputs follows from the network architecture)

The Landauer argument revisited

Onur Teker's argument is that Landauer's principle grounds information-priority physically: erasing a bit dissipates $k_B T \ln 2$ as heat, independently of any observer. This is correct for a *physical* computation.

But consider where the Landauer cost arises in a neural network running on a computer:

- **Not** in the abstract weight matrix (a mathematical object)
- **Not** in the network architecture (a geometric structure)
- **Yes** in the transistor switching operations on the physical processor — the substrate, not the network

The abstract neural network is geometrically defined and carries no Landauer cost. The physical instantiation carries Landauer cost. This is not a contradiction of Landauer's principle — it is a confirmation of the distinction between formal and physical information (cf. Doc. 022 and Doc. 230):

$$\Delta\text{CHSH}_{\text{formal}} = 0 \qquad \text{(keine physikalischen Kosten für abstrakte Distinktion)} \tag{243.8}$$

$$\Delta\text{CHSH}_{\text{elem}} = \frac{\xi}{2\pi} \approx 2 \times 10^{-5} \qquad \text{(Landauer cost per single physical measurement)} \tag{243.9}$$

Note: the cumulative deviation $\Delta\text{CHSH}(N) \approx 5\times10^{-4}$ at $N = 73$ runs (Doc. 022) is a *different* observable from the elementary cost $\xi/(2\pi)$ per measurement (Doc. 230). These are not in contradiction; they describe the same physical effect at different integration scales. The two values are consistent for $N \approx 2\pi/\xi \approx 47{,}000$ — the scale at which the cumulative formula reaches the elementary value.

The same distinction applies to neural networks: the geometry is formal; the Landauer cost arises when the geometry is physically instantiated.

Consequence for the priority debate

A neural network is simultaneously:

- An **information-processing system** at the functional level (Shannon)
- A **geometric structure** at the fundamental level (weight matrix, architecture, topology)

Neither description is more fundamental — they are projections of the same object at different levels of description. This is the structure of $\tilde{T}\cdot m = 1$: the same physical fact viewed from two sides. The geometry (weight matrix) is the condition; the information processing (input-output transformation) is the function. Neither generates the other; both are required for a complete description.

What AI systems show about the priority question: Even the most sophisticated AI — combining language models, memory, retrieval, and reasoning — is at its foundation a geometric structure (architecture, weights, connections) that is physically instantiated on a substrate. The information processing is *what the geometry does when instantiated*. The geometry does not derive from the information processing; the information processing derives from the geometry being run.

This does not resolve the ontological question of which is prior. It shows that the question is well-posed at every level of description — and that the geometric entry point is at least as natural as the informational one, since every AI system begins with a geometric design decision (architecture) before any information is processed.

.7 Numerical Verification: RNN Weight Matrix

The log-space network: geometry as a single weight

The FFGFT mass formula $m = r \cdot \xi^p \cdot v$ is nonlinear in p, so a standard linear network fails (errors of 50–2000 %). In log-space it becomes exactly linear:

$$\ln m = \underbrace{1}_{w_1} \cdot \ln r + \underbrace{\ln \xi}_{w_2} \cdot p + \underbrace{\ln v}_{b} \tag{243.11}$$

The analytic weights require no training: $w_1 = 1$, $w_2 = \ln \xi \approx -8.923$, $b = \ln v$.

Results

Particle	r	p	m_{FFGFT} [eV]	$\Delta\,\%$
Electron	$4/3$	$3/2$	5.054×10^5	1.09
Muon	$16/5$	1	1.050×10^8	0.57
Tau	$25/9$	$2/3$	1.785×10^9	0.46

Experimental values: $m_e = 5.110 \times 10^5$ eV, $m_\mu = 1.057 \times 10^8$ eV, $m_\tau = 1.777 \times 10^9$ eV.

What this shows

The weight $w_2 = \ln \xi$ *is* the T⁴ geometry — encoded as a single number. The network carries the *what* ($\xi = 4/30000$); it cannot carry the *why* (the winding topology that enforces this value). This confirms the projection π_I: functionally complete, geometrically incomplete.

.8 Quotient Structure: Why No Reverse Mapping

The mapping π_I is not injective. Different T⁴ states can map to the same information state:

$$(n_\theta, n_\varphi) \sim (k n_\theta, k n_\varphi) \quad \text{for any } k \in \mathbb{Z} \setminus \{0\} \tag{243.10}$$

Two winding configurations with the same *ratio* n_φ/n_θ produce the same spin $s = n_\varphi/(2n_\theta)$ and thus the same information state under π_I. The information formalism is therefore a **quotient theory** of FFGFT: it identifies configurations that are geometrically distinct.

This is not a deficiency — it is the precise sense in which the information formalism is a *coarser* description. The residual R_I is not a bug but a feature: it marks exactly what the quotient throws away.

Consequence for priority: An information formalism without geometry cannot decide *which* of the equivalent winding configurations is physically realised. The T⁴ geometry provides the **selection rule**: not all ratios n_φ/n_θ are admissible — only those satisfying the Z_3 topological constraint. The information formalism says “there are N distinguishable states”; the geometry says “these specific N are the admissible ones, and here is why”.

.9 Extension: Three AI Architectures as FFGFT Projections

Beyond RNNs, three fundamentally different architecture classes project FFGFT differently and leave different residuals.

1. Transformer (GPT, BERT, Llama)

Transformers use rotational position encodings (RoPE):

$$\text{RoPE: } \mathbf{q}_m = e^{im\theta} \cdot \mathbf{q}, \quad \theta \in [0, 2\pi) \tag{243.12}$$

The phase θ_4 can be encoded as relative position — but as a fixed encoding, not a dynamic variable. Residual: ε is lost; ξ is learned as attention temperature, not as winding ratio.

2. Modern Hopfield Network (Demircigil et al. 2017)

The modern Hopfield network has an exponential energy function with inverse temperature parameter $\beta = 1/T$:

$$E = -\frac{1}{\beta} \log\left(\sum_{\mu} e^{\beta \cdot \text{sim}(\xi, \xi_\mu)} \right) \tag{243.13}$$

For finite β, ε appears as a structural parameter, not training noise. **This is the best information-theoretic approximation of ε so far.** Residual: θ_4 and individual winding numbers are still lost.

3. Graph Neural Networks (GNNs)

Winding numbers (n_θ, n_φ) can be encoded as weighted nodes. Residual: θ_4 and ε are lost.

Summary of three architectures

Architecture	(n_θ, n_φ)**?**	θ_4**?**	ε**?**
RNN (real)	Partial	No	No
Transformer (RoPE)	Indirect	Partial (fixed)	No
Hopfield (modern)	No	No	**Yes** (as β)
GNN	Yes (nodes)	No	No
No architecture encodes all three simultaneously.			

.10 The Inverse Problem: Can AI Reconstruct FFGFT?

No-free-lunch for topological invariants: From finitely many bits, the complete T^4 geometry cannot be uniquely reconstructed:

$$\nexists \text{ AI, reconstructing } (n_\theta, n_\varphi, \theta_4, \varepsilon) \text{ uniquely from } \{I, \Theta(\theta_4), \langle\varepsilon\rangle\} \tag{243.14}$$

Current research (representation learning, persistent homology, contrastive learning) can approximate individual aspects but not the specific number $\xi = 4/30000$ — it must be set as a hyperparameter and does not follow from information alone.

.11 Testable Prediction: AI Hardware as T^4

Every physical AI system runs on a processor with:

- A clock (cyclic time: θ_4 as clock phase)
- Finite registers (discrete states: $\log_2 N$ bits)
- Thermal noise (ε as jitter)

FFGFT prediction: Torus-networked chips show lower Landauer costs than 2D-grid chips:

Chip topology	τ	Exp. cost	FFGFT prediction
2D-Grid (standard)	0	$k_B T \ln 2$	higher than ξ
2D-Torus (TPU)	0.5	$\approx 0.67 \cdot k_B T \ln 2$	between ξ and standard
3D-Torus (Cerebras)	0.8	$\approx 0.33 \cdot k_B T \ln 2$	closer to ξ
4D-Torus (optical)	1.0	$\xi \cdot m_e c^2 / \ln 2 = k_B T_{\text{CMB}}$	exactly ξ

Experimental decision of the priority debate:

- If costs depend on chip topology: geometry (torus) is fundamental, information follows from it.
- If all topologies show equal costs: information is fundamental.

FFGFT's prediction is clear: **Torus topologies win.**

Notation: Chip Topology Parameters

Symbol	Meaning
τ	Torus similarity of chip topology ($\tau = 0$: 2D grid, $\tau = 1$: ideal T^4)
α	Material-dependent coupling factor (photon-phonon vs. Coulomb)
OCS	Optical Circuit Switch (MEMS-based)
ICI	Inter-Chip Interconnect (Google proprietary high-speed protocol)
WSE	Wafer-Scale Engine (Cerebras)
T^4	4-torus: four compact loops (FFGFT full geometry)
T^3	3-torus: three compact loops (TPUv4 physical topology)
RoPE	Rotational Position Encoding (Transformer position encoding)
β	Inverse temperature parameter in modern Hopfield network ($\beta = 1/T$)
π_{elec}	Projection onto electrical mesh topology (additional residual)

Clarification: Photonic vs. Electrical Chip Topologies

Not all torus chips are photonic. The following overview distinguishes between logical and physical topology realisation, based exclusively on verified sources.

System	Topology	Phot.?	Source / Note
Google TPUv4 Pod	3D Torus ($4\times4\times4$ cube)	**Yes** (OCS)	Jouppi et al. 2023, arXiv:2304.01433 Palomar OCS: MEMS mirrors, 136×136 ports Wrap-around via optical fibre, reconfigurable in 10 s
Google TPUv5p	3D Torus ($16\times20\times28$)	**Yes** (OCS)	13,824 optical ports, 48 OCS units (based on Google documentation)
Morphlux (Cornell/Lightmatter)	Reconfigurable (T^3 emulable)	**Yes** (silicon photonics)	Kumar et al. 2025, arXiv:2508.03674 Programmable photonic chip-to-chip interposer 1.72× training throughput improvement
Cerebras WSE-3	2D Mesh (not a torus)	No (electrical)	Cerebras SDK documentation; sdk.cerebras.net 900,000 cores, 2D rectangular mesh on one wafer
Standard GPU cluster	2D Grid / Fat-Tree	No (electrical)	NVLink / InfiniBand between chips

Correction: Cerebras WSE-3 is not a torus but a **2D mesh** — topologically related but without the periodic boundary conditions that define a torus. The FFGFT prediction (lower Landauer costs for torus topologies) therefore applies to TPUv4/v5p, not to the Cerebras WSE-3.

Why photonic torus chips are particularly relevant for FFGFT

In the TPUv4, the optical circuit switch (OCS) realises the torus geometry *physically*: a light signal leaving the torus at one edge re-enters at the opposite edge — exactly like the T^4 boundary condition. The light "lives" on the torus.

In electrical mesh topologies (Cerebras WSE-3), the connection is defined logically by routing tables, not by the physical propagation of signals. This corresponds to a projection π_{elec} with additional residuals compared to the ideal torus.

Consequence: The proposed experiment (table of τ vs. Landauer costs in Doc. 243 Section 12) should treat photonic torus chips (TPUv4/v5p) and electrical mesh chips (Cerebras WSE-3) separately. The limit $\tau \to 1$ is only expected for photonic 4D tori.

.12 Open Points and Next Steps

Answered by Calculation

- **Numerical verification (answered):** The log-space network with a single weight $w_2 = \ln\xi \approx -8.923$ reproduces all three lepton masses without training: electron $\Delta = 1.09\,\%$, muon $\Delta = 0.57\,\%$, tau $\Delta = 0.46\,\%$. The network is π_I in pure form: functionally complete, geometrically incomplete.
- **Constraint $T^+ + T^- = 0$ (partially answered):** An antisymmetric weight matrix $W = -W^\top$ can encode this condition structurally. For state $h = (h^+, h^-)$ with $h^+ + h^- = 0$, the condition is preserved as a conservation law when W is antisymmetric and h is antisymmetrically initialised. However, this requires a special architecture (antisymmetric weights) and is not present in a standard RNN.

Still Open

- **Formal derivation** of the network architecture from the T^4 projection geometry (analogous to Doc. 189 for spinors)
- **Coefficient** $275/4$ in the T_{CMB} correction: $T_{\text{CMB}} = \frac{16}{9}\xi(1-\frac{275}{4}\xi) = 2.725486$ K ($\Delta = 0.0002\,\%$). The tetrahedral number $68 = 72 - 4$ (Doc. 003) gives $\Delta = 0.01\,\%$, the number 69 gives $\Delta = 0.003\,\%$. The exact value $68.75 = 275/4$ has no geometric derivation yet.
- **Connection to UIFT**: If ξ is measured thermodynamically (Vopson-Teker), this confirms the bridge — but not the priority direction. Whether ξ is primarily geometric or thermodynamic remains undecided by this test.

Literature (Chip Topologies)

- **Google TPUv4:** Jouppi, N. P. et al. (2023). TPU v4: An Optically Reconfigurable Supercomputer for Machine Learning with Hardware Support for Embeddings. *Proc. ISCA 2023*. arXiv:2304.01433. Quote: *"Optical circuit switches (OCSes) dynamically reconfigure its interconnect topology [...]; users can pick a twisted 3D torus topology if desired."*
- **Morphlux:** Kumar, A. V., Ding, E., Devraj, A., Bunandar, D., & Singh, R. (2025). Morphlux: Transforming Torus Fabrics for Efficient Multi-tenant ML. arXiv:2508.03674. Accepted at ASPLOS 2026.
- **Cerebras WSE-3:** Cerebras Systems. *Cerebras SDK Documentation — A Conceptual View*. sdk.cerebras.net. Quote: *"The PEs are interconnected by communication links into a two-dimensional rectangular mesh on one single silicon wafer."*

Internal Document References (FFGFT Series)

The following internal documents are cited in the text. They are available in the GitHub repository: `github.com/jpascher/T0-Time-Mass-Duality`

Doc.	Title	Content (relevant for Doc. 243)
009	Origin of ξ	$K = 245$, $\kappa = 7$, e-p-μ triangle
133	Fractal Correction Derivation	$K_{\text{frak}} = 1 - 100\xi$, RG run
189	Spinor Winding Structure	(n_θ, n_φ) mapping to spin
190	Corrections Register	R10–R12: $K = 245$, T_{CMB}, CHSH
193	Non-closure ε	$\varepsilon \approx 0.001$ as arrow of time
206	Triangle-Matrix Reduction	Geometry generates matrix, not vice versa
230	Hilbert Space Translation	Bijective FFGFT ↔ QM mapping
231	Hilbert Space Extension	Bell states, CHSH from FFGFT

Key Result

Core statement of Doc. 243:
FFGFT is mappable into the information formalism (as projection π_I), analogous to the mapping into Hilbert space (Doc. 230). In the mapping, the θ_4 phase, individual winding numbers, and the geometric origin of the non-closure are lost. The parameter ξ is invariant under π_I and serves as bridge between geometric and thermodynamic description. Mappability does not prove ontological priority of information — it shows that both descriptions are projections of the same structure.